

Assessment of perceived indoor environmental quality, stress and productivity based on environmental sensor data and personality categorization

Johanna Kallio^{a,*}, Elena Vildjiounaite^a, Jani Koivusaari^a, Pauli Räsänen^a, Heidi Similä^a, Vesa Kyllönen^a, Salla Muuraiskangas^a, Jussi Ronkainen^a, Jari Rehu^a, Kaisa Vehmas^b

^a VTT Technical Research Centre of Finland Ltd., Kaitoväylä 1, 90571, Oulu, Finland

^b VTT Technical Research Centre of Finland Ltd., Vuorimiehentie 3, 02150, Espoo, Finland

ARTICLE INFO

Keywords:

Indoor environmental quality
Stress
Productivity
Classification
Machine learning
Personality

ABSTRACT

Indoor environmental quality (IEQ) has an influence on peoples' health, cognitive performance and productivity in school and office environments. This study used environmental temperature, humidity, air pressure, and CO₂ sensor data collected during 3.5–7 months in an office and a school facility to classify occupants' perceptions of IEQ, stress and productivity in two classes, "negative" and "positive". Self-reported data from 15 office workers and four teachers were used to train person-specific SVM classifier models. Relatively high accuracies were achieved in classifying IEQ (84%), stress (88%) and productivity (92%) using different combinations of environmental sensor data. Furthermore, the associations between the Big Five personality trait variables (neuroticism, extraversion, openness, agreeableness and conscientiousness) and negative experiences regarding stress, productivity and IEQ were investigated. Positive correlation was found between extroversion and co-occurring stress and IEQ problems, which suggest that more extroverted people more likely to be stressed by insufficient environmental quality or to be more sensitive to environmental factors when under stress. Overall, the results indicate that it is possible to measure and classify perceived IEQ, stress and productivity sufficiently accurately using inexpensive environmental sensors.

1. Introduction and related work

During the past few decades increasing evidence has been accumulated on the fundamental impact of indoor environmental quality (IEQ) on health and performance. IEQ refers to the interaction between physical, chemical and biological factors indoors, such as air quality, temperature, relative humidity, luminosity and sound level, and their influence on quality of life [1]. These environmental factors, known as environmental stressors, can cause distress by distracting an individual's attention [2]. Despite increased awareness about the importance of IEQ, there is still much room for improvement. Demand for high-quality IEQ solutions has seen a growing trend in recent years, especially in schools, workplaces, and public spaces. For example, energy regulations for the building and construction industry have been tightened to reduce greenhouse emissions and energy costs in general. Moreover, as a drive

for construction cost-efficiency and real estate profit return, smaller spaces are used to accommodate more people. As a response to the stringent regulations, buildings have been constructed to be more energy efficient and tighter and, as a side effect, decreased ventilation together with emissions from building materials have contributed negatively to IEQ [3]. Furthermore, even in the context of smart buildings, the needs of building users are often ignored [4], even though motivated and healthy employees are emerging as a strategic priority of enterprises [5].

It has been estimated that 80–90% of business operating costs are attributable to employee expenses, whereas only 1–3% are associated with energy costs and 8–11% with other facility costs [6,7]. Moreover, it has been calculated that the influence of poor IEQ on productivity, and consequently on employee expenses, can exceed energy costs by two orders of magnitude [8]. Improving IEQ can lead to increased health and

* Corresponding author.

E-mail addresses: johanna.kallio@vtt.fi (J. Kallio), elena.vildjiounaite@vtt.fi (E. Vildjiounaite), jani.koivusaari@vtt.fi (J. Koivusaari), pauli.rasanen@vtt.fi (P. Räsänen), heidi.simila@vtt.fi (H. Similä), vesa.kyllonen@vtt.fi (V. Kyllönen), salla.muuraiskangas@vtt.fi (S. Muuraiskangas), jussi.ronkainen@vtt.fi (J. Ronkainen), jari.rehu@vtt.fi (J. Rehu), kaisa.vehmas@vtt.fi (K. Vehmas).

<https://doi.org/10.1016/j.buildenv.2020.106787>

Received 27 December 2019; Received in revised form 10 February 2020; Accepted 3 March 2020

Available online 10 March 2020

0360-1323/© 2020 The Author(s).

Published by Elsevier Ltd.

This is an open access article under the CC BY-NC-ND license

(<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

productivity and therefore more profitable investments with respect to employee expenses. For instance, evaluations of financial investments in IEQ supporting health and productivity have shown to be cost effective, with a payback time of less than two years in general [9].

1.1. IEQ, productivity and stress

Various laboratory and real-life studies have reported associations between ventilation rate and IEQ measures, particularly in school and office environments (e.g. Ref. [10,11]). Moreover, associations between insufficient IEQ and impaired health, cognitive performance and productivity are well established. Wargocki and Wyon [8] summarized the results of several laboratory and field studies investigating the influence of indoor air quality and thermal environment on performance, such text-typing, reading, memorization and mathematical tasks of the test subjects. The reviewed studies demonstrated that increased ventilation rate and thermal comfort improve the performance of office workers and schoolchildren. In addition, although air quality and temperature are typically accepted by the majority of building occupants, personal control over the thermal environment has been suggested to achieve optimal work performance as discomfort can decrease performance by up to 10% for office workers and up to 30% for schoolchildren [8]. Another 8-month field study in office environments [12] showed that perceived thermal discomfort, noise annoyance and lighting discomfort reduced work performance and objectively measured cognitive performance by between 2.4% and 14.8%. However, environmental stressors affected work performance indirectly, for example, by reducing motivation and focus. In addition, the number of environmental stressors was found to be associated with a nearly linear reduction in self-reported work performance, indicating the cumulative effect of environmental stressors [11].

Al Horr et al. [13] recapitulated in their review article that thermal comfort, indoor air quality, office layout, and noise and acoustics play a significant part in office worker productivity, and dissatisfaction with these environmental factors can lead to productivity losses. Another study [14] suggested that the thermal environment also has an influence on the perceptions of other IEQ factors, such as air quality, noise and lighting. When the test subjects experienced the thermal environment as unsatisfactory, they also reported dissatisfaction with other IEQ factors, whereas increased thermal satisfaction had a positive influence on productivity. Best productivity was achieved when thermal environment was perceived as “neutral” or “slightly cool” [14]. However, research on the relationship between office tasks, temperature and productivity has shown that different kinds of mental work may have different optimal temperatures for improved productivity [13]. Field studies typically employ subjective measures of productivity, such as self-reports of employees, but the subjective feelings of the test subject have not corresponded to objective productivity measures in the studies comparing these two approaches [8]. On the other hand, the discrepancy between these two approaches has been inconsistent [8].

Laboratory conditions do not accurately represent actual working conditions, because in lab studies environmental or psychosocial stress is usually induced by giving test subjects relatively short tasks or adjusting a single environmental factor (e.g., temperature) while researchers observe the ability of test subjects to perform certain tasks. The lab studies are relatively short, varying from a few hours to one day, and typically measure short periods of intense conditions. In real life, people behave in a greater variety of ways than during lab tests, and environmental or psychosocial stress can be caused by several different factors. Hence, findings from lab studies cannot be directly extrapolated in real life, and further studies with objective measures and long-term assessments in real environments are required. Kumar et al. [15] highlighted the importance of real-time sensing and quantitative data to provide more accurate spatial and temporal data on air quality for more effective interventions. According to a large-scale study [10], seasonal monitoring is also recommended due to significant seasonal variations

in measured air quality and thermal comfort data.

Even though physiological sensors have demonstrated high potential for objective stress detection in lab studies, different types of physical activities impair the quality of data in real-life measurement [16]. For this reason, the majority of real-life stress recognition studies have been based on behavioral data gathered from smartphones (e.g. Refs. [17–20]). Using mobile phones as the only data source does not induce extra disturbance for users, but continuous data collection may rapidly drain phone batteries [21]. Using environmental sensors, instead, can overcome the problems with smartphone battery life and maintenance effort. However, only a few studies have used environmental sensors for stress detection in real life to date. Examples of previous studies include utilization of audio data [22], keyboard/mouse data [23] and depth camera data [24], but it should be noted that especially audio analysis and keyboard/mouse usage logging may raise privacy concerns. On the contrary, some environmental sensors, such as temperature, humidity, air pressure or carbon dioxide (CO₂), are less concerning from the privacy point of view. A previous study on employing environmental data for assessing stress and indoor conditions reported reasonable classification accuracies, but the study had only four test persons, all of them female, and the sampling frequency of environmental data was only once every 15 min, which is rather sparse when measuring environmental changes [25].

1.2. Individual characteristics and person-specific learning

Providing a well-being and productivity promoting working environment is further complicated by the fact that people are individuals and occupant comfort depends on various independent personal variables, such as individual metabolism, clothing preference, and activity patterns [13]. Personal sensation of environmental comfort is affected by physiological and physical characteristics, but individual expectations, environmental context, and the availability of control can also influence personal perceptions of environmental acceptability [26]. For instance Ref. [27], found only a weak association between perceived thermal comfort and measured IEQ (temperature, relative humidity, and CO₂) parameters. Personality has been shown to affect exposure to stressors, influencing the nature and severity of stressor experiences [28]. Personality can be seen as an underlying behavioral and emotional description of how the individual feels and acts on average, and what kind of physiological and behavioral reactions arise. Currently, a widely applied model for describing personality is the Big Five model [29], which consists of five global factors of neuroticism (N), extroversion (E), openness to experience (O), agreeableness (A) and conscientiousness (C). In general, high scores of neuroticism are linked with negative emotions, such as anger and anxiousness; high scores of extroversion are linked with socializing and impulsiveness; high scores of openness are linked with curiosity; high scores of agreeableness are linked with trustworthiness and kindness; and high scores of conscientiousness are linked with self-discipline and vigilance [29]. Therefore, stress and environmental comfort are highly subjective experiences and some individuals are more vulnerable to specific psychosocial or environmental stressors, which may strain wellbeing and performance [30,31].

Several previous lab (e.g. Ref. [32,33]) and field (e.g. Ref. [17,18]) studies on stress recognition have reported notably higher accuracies with person-specific stress classifiers than general stress classifiers trained based on the data of many users. Average differences between the accuracies of person-specific and general approaches exceed 10–20%. We employ a supervised classifier, because the majority of existing stress classifiers [34], as well as thermal comfort models [35], are trained using fully supervised learning. More specifically, we use Support Vector Machines (SVMs), one of the most popular methods of learning from relatively small datasets (person-specific data are typically not abundant, at around 50–100 self-reports per person). SVMs have been successfully demonstrated in many studies measuring stress [34] or environmental satisfaction [36,37].

1.3. Objectives

Analysis of long-term real-life IEQ data is needed to understand environmental stressors and their person dependency. In this study, we use a person-specific data analysis approach to identify personal perceptions of IEQ, stress and productivity based on environmental sensor data. The main contribution of this work is the following. First, we introduce a privacy-safe and inexpensive setup for long-term collection of environmental sensor data (temperature, humidity, air pressure, and CO₂) from real office and school environments. Secondly, we present person-specific data analysis results from 3.5-month and 7-month real-life pilots using objective IEQ data to assess personal perceptions of four teachers and 15 office workers regarding “negative” or “positive” IEQ, stress and productivity. The achieved classification accuracies confirm the feasibility of using objective IEQ data to assess personal perceptions of IEQ, stress and productivity. Thirdly, this work compares how personality trait variables (N, E, O, A, C) and personal perceptions of IEQ, stress and productivity are associated. The results indicate that personality traits may have an influence on expressing or experiencing environmental comfort.

2. Methods

2.1. Recruitment and study participants

The aim of this study was to assess occupants’ perceived IEQ, stress and productivity using low-cost sensors and machine learning, and compare the analysis results of two different types of facilities. Considering that children spend the majority of their time in school and a large proportion of adults in office facilities [38], we recruited participants from an elementary school and from a knowledge worker office located in Northern Finland. Inclusion criteria for the test subjects were a) Android phone in use, b) commits to wear wrist-devices for data collection during the workday, and c) commits to answer self-reports daily. We targeted only Android phone users because the application used for collecting self-reported data was developed for the Android OS (described in more detail in “Collection of self-reporting data” in Section 2.3). The test subjects wore a wrist device to monitor physiological data (continuous activity and heart rate) during working hours. However, the physiological data was not used in the analyses reported in this paper.

There were a total of 27 voluntary and eligible test subjects (4 teachers, 23 office workers). The average age of the voluntary teachers was 43.5 years (standard deviation 15.6) and the office workers 41.6 years (standard deviation 8.0). All (4/4) of the teachers and 11/23 of the office workers were females. Information about the study and informed consent forms were delivered to the teachers in a face-to-face group meeting and to the office workers individually. The applications (self-report and physiological monitoring) were installed for the teachers during the session; the office workers installed the self-report applications independently. After the participants gave written informed consent, they filled in a pre-study questionnaire (background information, symptoms, perceived air quality, perceived stress), started wearing the wrist-device, and started using the self-report Android app. At the end of the study, the participants filled in an abbreviated form of the Big Five Inventory-2 (BFI-2-S) questionnaire [39]. The BFI-2-S questionnaire includes 15 items and covers the “big five” sub-factors of neuroticism, extroversion, openness to experience, agreeableness, and conscientiousness. Responses to the questions are provided with a 5-point Likert scale from “strongly disagree” (1) to “strongly agree” (5). All participants responded to the questionnaires in Finnish. The study protocol was reviewed and accepted by the Ethics Review Board of VTT Technical Research Centre of Finland.

2.2. Data collection

The main objectives for the data collection were: (1) to collect

continuous and objective IEQ sensor data, and subjective perceptions of the test subjects during several months; and (2) to collect objective and subjective data in two different types of working environments: first, where occupants have reported dissatisfaction with IEQ (school facility) and second, where occupants have not reported dissatisfaction with IEQ (office facility). The data collection period was 7 months for the office and 3.5 months for the school facility.

2.2.1. Self-reporting data

The test subjects reported their perceptions of IEQ, stress, productivity and possible health symptoms via an Android self-reporting application. Fig. 1 presents screenshots of the self-reporting application. Details on implementation of the self-reporting application and preliminary analysis of the self-reporting data from the school pilot have been reported in Refs. [40]. After the user had logged in with a user-specific pseudo-anonymization code at the beginning of the study, the application reminded her/him to answer questionnaires at given times of day on weekdays. Two time-triggered questionnaires were scheduled per working day (morning and afternoon); the implementation enabled personalization of the times when each questionnaire prompt was delivered. The schedule allowed the test subjects to choose a convenient moment to answer, but reduced the probability of collecting outdated answers.

The morning questionnaire asked about how rested the test subject felt and perceived IEQ. Perceived IEQ was queried with one general question and five additional items about air freshness, odor intensity, temperature, humidity, and noise level on a 5-point answer scale. The afternoon questionnaire asked the user about their perception of IEQ again and, additionally, their perceived productivity and stress. The user was also asked which room they were currently in and about any possible symptoms related to insufficient IEQ. The reportable symptoms were related to head, nose, throat, breathing, eyes, skin, nausea or tiredness. In addition, the application provided a button for on-demand reporting that allowed each user to submit a self-report on a chosen topic at any time, also on 5-point scale. The reported answers were stored in MongoDB as one document per questionnaire item (i.e., one question) along with a pseudo-anonymization code. The answers were retrieved via REST API. The specific questions used in the analyses of this paper are presented in Table 1.

2.2.2. Environmental sensor data

Environmental sensor data was collected using IoT-based monitoring systems in both piloting premises, an elementary school and an office facility. The monitoring systems were installed in four classrooms and 12 office rooms. All piloting rooms were equipped with a set of customized sensor devices based on the VTT Tiny Node hardware platform with commercially available sensors (THP-CO₂, THP-DOOR and THP-PIR) (see Table 2). Tiny Node [41] is a sensor device of euro coin size, operated by cell battery and communicating wirelessly over Bluetooth Low Energy (BLE). The node has built-in flexibly configurable signal processing pipelines and interfaces for several types of sensors including but not limited to acceleration, temperature, humidity and pressure sensors.

The measured IEQ parameters included temperature (T), relative humidity (H), air pressure (P) and carbon dioxide (CO₂). Additional measurements included presence data via passive infrared (PIR) sensors and door state data (open or closed) via magnetic latch sensors (DOOR). In this paper, however, PIR and door latch data were not included in the analyses. Table 2 lists the sensor models utilized in the Tiny Node based sensor devices. The sensor accuracies of the customized devices were manufacturer calibrated, and only an informal cross-check was carried out for this study. Prior to deployment in the test environment, all of the sensors were placed in the same space at the same time to cross-check that the sensors show consistent readings within the sensor manufacturer’s specified accuracy range. The cross-check was performed over one week in an office room setup similar to that depicted in Fig. 2.

The figure displays four screenshots of a mobile application interface, arranged in a 2x2 grid. Each screenshot shows a different section of the app, all with a dark blue header and a light blue body.

- Top Left: Indoor air quality**
 - Header: Indoor air quality
 - Question: How would you estimate the at the moment **freshness of air?**
 - Options:
 - ☐ Extremely stuffy
 - ☒ Stuffy
 - ☐ Neutral
 - ☐ Fresh
 - ☐ Extremely fresh
 - Footer: Describe more details if you wish.
- Top Right: Indoor air quality**
 - Header: Indoor air quality
 - Question: How would you estimate the at the moment **temperature?**
 - Options:
 - ☐ Cold
 - ☒ Cool
 - ☐ Neutral
 - ☐ Warm
 - ☐ Hot
 - Footer: Describe more details if you wish.
- Bottom Left: Stress**
 - Header: Stress
 - Question: There is positive and negative **stress**. What kind of stress you felt **today?**
 - Text:

Positive stress = Enthusiasm, excitement, inspiration due to current or recent challenge (e.g. the feeling when you successfully solve a difficult problem).

Negative stress = distress, nervousness or anxiety due to the current or recent challenge (e.g. fear of failure)
 - Options:
 - ☐ Strong positive stress
 - ☐ Positive stress
 - ☐ Neutral
 - ☐ Negative stress
 - ☐ Strong negative stress
- Bottom Right: Productivity**
 - Header: Productivity
 - Question: How productively did you work today?
 - Options:
 - ☐ Much more productively than usually
 - ☐ More productively than usually
 - ☐ As usually
 - ☐ Less productively than usually
 - ☐ Much less productively than usually

Fig. 1. Screenshots of air freshness, thermal comfort, stress, and productivity related questions in the self-reporting application.

All of the sensor devices were positioned according to national legislative recommendations [42], i.e., at about 1.1 m above floor level, excluding door status sensors placed above door height. In the office cubicles, the PIR sensors were positioned close (1–2 m) to the seating position, such that each sensor monitored a single target person only. In the classrooms, multiple PIR sensors were positioned such that each sensor covered the maximum possible area without overlapping with

adjacent PIR sensors. Fig. 2 shows the typical sensor positioning in offices and classrooms. The overall building of the test environment has a balanced mechanical ventilation system for both inlet and outlet flow adjusted and calibrated to zero pressure difference. Inlet and outlet valves were located at ceiling level at the centre and edges of the room, respectively. The inlet valves were of a diffuser type and had been observed to mix the air efficiently in the room. For the target of

Table 1

Self-reporting application questions used in the analyses.

Question	Answer alternatives on a 5-point scale				
Overall IEQ perception: How would you estimate the indoor environment quality at the moment? *) Indoor environmental quality is affected by oxygen level, odors, temperature, noise and humidity	Very bad	Bad	Acceptable	Good	Very good
Air freshness: How would you estimate air freshness at the moment?	Very bad	Bad	Acceptable	Good	Very good
Odor: How would you estimate the odor intensity at the moment?	Unbearable odor	Strong odor	Moderate odor	Slight odor	No Odor
Thermal comfort: How would you estimate the temperature at the moment?	Hot	Warm	Neutral	Cool	Cold
Stress (perception of own stress): Stress can be measured as positive or negative. What kind of stress you felt today? Positive stress = Enthusiasm, excitement, determination, inspiration due to current or recent challenge (e.g. the feeling when you successfully solve a difficult problem). Negative stress = distress, nervousness or anxiety due to a current or recent challenge (e.g. fear of failure)	Strong negative stress	Negative stress	Neutral	Positive stress	Strong positive stress
Productivity (perception of own productivity): How productively did you work today?	Much less productively than usually	Less productively than usually	As usual	More productively than usually	Much more productively than usually

Table 2

Sensor devices and variables of the IoT monitoring system.

Sensor variables	Sensor model	Measurement range (accuracy)	Sample interval
Temperature [°C]	Silicon Labs Si7021	−10...85 °C (±0.4 °C)	60 s
Relative humidity [RH-%]	Silicon Labs Si7021	0 ... 80% (±3%)	
Air pressure [hPa]	Bosch BMP180	300...1100 hPa (−4 ...+ 2 hPa)	
CO ₂ [ppm]	SENSEAIR K30	0...5000 ppm (±30 ppm ± 3% of reading)	
PIR count [0...12]	Panasonic EKMB1301113K	Sum of movement events detected at 1/5 Hz over a time window of a minute open/closed	
Door state [0 = door open, 1 = door closed]			

capturing typical CO₂ concentration and its variations in the room, CO₂ sensor devices were placed closer to the ventilation outlet than the inlets, as depicted in Fig. 3 below.

Data from Tiny Nodes was obtained once per minute. The measurement infrastructures at the pilot sites included commercial gateway devices for communicating with the sensor devices and aggregating their data. The gateway devices in all school pilot rooms utilized a single commercial router device with 4G mobile network capabilities for internet access, whereas the gateway devices in the office rooms utilized available wireless or wired internet connection. All gateways were run on a Raspberry Pi single-board computer employing the Raspbian operating system (a Linux variant), and data was sent to the Microsoft Azure cloud platform via MQTT protocol over TCP/IP. Data retrieval from Azure utilized a published API for Microsoft Azure TableStorage. We implemented a client application in Python language and based on Cosmos DB API, which provided an interface to TableStorage via a SW module called TableService [43]. Data were retrieved periodically in batches.

2.3. Data analysis

Since the interpretation of IEQ depends on the individual, we conducted a user-dependent data analysis. To evaluate the conditions of each person, a classifier was trained by using data specific to each person. The notion of “condition” is used here to refer to dissatisfaction with IEQ, stress, and own productivity. The results of assessing these

conditions for different groups, teachers and office workers, are presented in Section 3.2 “Classification of office pilot data” and Section 3.3 “Classification of school pilot data”.

The selected machine learning approach is supervised learning, which is commonly applied in stress detection and assessment of satisfaction with environmental conditions. In the analysis of IEQ data, we used a SVM implementation in the Scikit-learn Python library [44]. Subjective perceptions of the test persons regarding their conditions were used as class labels.

We also used the tsfresh Python library [44] for feature extraction. Features were extracted from time series data for time windows of pre-defined length (2 h). The Tsfresh library extracts over 300 features from the time series data, such as minimum/maximum/median values of the samples within the time window, absolute energy of the time series (the sum over the squared values), autocorrelation, Fast Fourier Transform coefficients, variance, quantiles, skewness, kurtosis, etc. A full list of the extracted features is given in Ref. [45]. The Tsfresh library uses all samples in the time window for feature calculation, but if many samples in the window are missing, the feature values become untrustworthy. Therefore, in this study we ignored time windows with more than 20% missing samples. After feature extraction, we performed feature selection by calculating correlations between all extracted feature values and respective labels and selecting the features with the highest correlation. This approach enables selecting the most suitable features for each problem individually. As a result, the number of selected features varied between test subjects and questions.

2.3.1. Data classes

We conducted a two-class classification in which the SVM was trained to identify between “bad” conditions and “OK” conditions. In IEQ comfort identification, the “bad” class included “bad” and “very bad” answers, whereas the “OK” class included all other answers. For stress detection, the “bad” class included “negative” and “very negative” self-reports, and the “OK” class included “neutral”, “positive” and “very positive” self-reports. Similarly, for productivity classification the “bad” class included “much less productively than usual” and “less productively than usual” answers, while the “OK” class included all other answers. The SVM classifier was then trained to map sensor data into one of the two selected classes of individual perception, for instance, to learn whether self-report for stress is in the “bad” class or “OK” class.

2.3.2. Experimental protocol

Due to the size of the training data, we used a leave-one-self-report-out protocol for each test subject. Each self-report and the corresponding

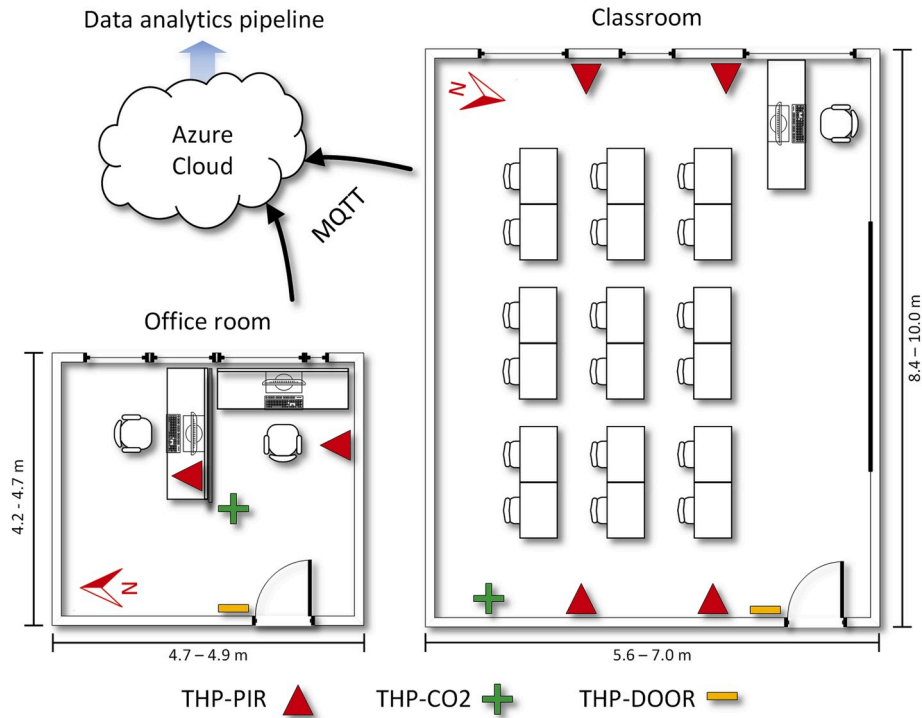


Fig. 2. Data acquisition system overview.

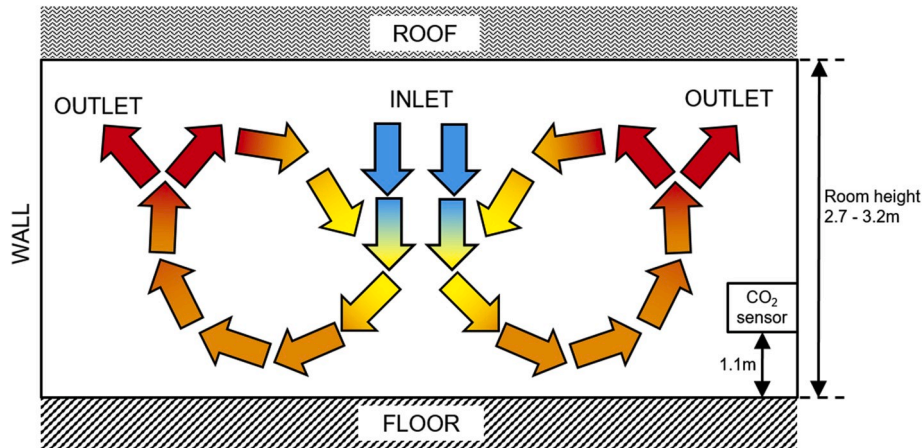


Fig. 3. Sensor, including CO2, positioning in a room.

sensor data (average from previous 2 h) were selected as test data, and all remaining data were used for feature selection and SVM training. This process was repeated for all self-reports of the test subject, and the same procedure was repeated for each of the test subjects.

Several IEQ-related questions were combined together as follows:

- **Any IEQ problem:** If a test subject provided a “bad” or “very bad” answer to any of the four IEQ questions (**Overall IEQ perception, Air freshness, Odor, Thermal comfort**), then **Any IEQ problem** belongs to the “bad” class. Otherwise, it belongs to the “OK” class.

For each subject, a separate SVM model was trained and tested for each of the following problems: **Overall IEQ perception, Air freshness, Odor, Thermal comfort, Any IEQ problem, Stress** and **Productivity**. For each model, sensor data features were extracted from the time window prior to each self-report in the training data. Time window length was set at 2 h. However, the problem of determining the optimal time

window needs further work. Then, correlations between features and perceptions of the test subjects were calculated from the training data, and the SVM models were trained.

For the testing phase, the same features were extracted from the test data, each test sample was classified with an SVM, and the SVM output was compared with the corresponding self-report. Classification accuracy was estimated according to the following equations:

$$\text{Total Accuracy} = \frac{N_{\text{correctBad}} + N_{\text{correctOK}}}{N_{\text{bad}} + N_{\text{OK}}} \quad (1)$$

$$\text{True OK} = \frac{N_{\text{correctOK}}}{N_{\text{OK}}} \quad (2)$$

$$\text{True Bad} = \frac{N_{\text{correctBad}}}{N_{\text{bad}}} \quad (3)$$

In equations (1)–(3), $N_{\text{correctBad}}$ is the number of correctly classified “bad” answers; $N_{\text{correctOK}}$ is the number of correctly classified “OK”

answers, and N_{bad} and N_{OK} are the numbers of “bad” and “OK” answers, respectively.

2.3.3. Correlations between personality variables and self-reports

To understand better how personality is associated with perceptions of IEQ, stress and productivity, we investigated correlations between five personality variables (N, E, O, A, C) and ratios of different “bad” self-reports of 16 test subjects who responded to the Big Five questionnaire. First, we calculated the ratios of different “bad” self-reports to all self-reports for the responded 16 test subjects with following equations:

$$IEQ \text{ ratio} = \frac{N_{IEQBad}}{N_{days}} \quad (4)$$

$$Stress \text{ ratio} = \frac{N_{StressBad}}{N_{days}} \quad (5)$$

$$Productivity \text{ ratio} = \frac{N_{ProBad}}{N_{days}} \quad (6)$$

In equations (4)–(6), N_{IEQBad} is the number of days when “bad” IEQ was, $N_{StressBad}$ is the number of days when “bad” stress was reported, N_{ProBad} is the number of days when “bad” productivity was reported and N_{Days} is the number of days when self-reports were provided.

We also studied the correlations between personality variables and co-occurring “bad” self-reports of these persons. For this purpose, we calculated ratios for co-occurring “bad” self-reports for the responded 16 test subjects with following equations:

$$Stress \text{ Productivity ratio} = \frac{N_{StressBad_and_ProBad_on_same_day}}{N_{ProBad}} \quad (7)$$

$$IEQ \text{ Stress ratio} = \frac{N_{IEQBad_and_StressBad_on_same_day}}{N_{StressBad}} \quad (8)$$

$$IEQ \text{ Productivity ratio} = \frac{N_{IEQBad_ProBad_on_same_day}}{N_{ProBad}} \quad (9)$$

In equations (7)–(9) $N_{StressBad_and_ProBad_on_same_day}$ is the number of days when “bad” stress and “bad” productivity were reported together, and N_{ProBad} is the number of days when “bad” productivity was reported; $N_{IEQBad_and_StressBad_on_same_day}$ is the number of days when “bad” IEQ and “bad” stress were reported together, and $N_{StressBad}$ is the number of days when “bad” stress was reported; and $N_{IEQBad_and_ProBad_on_same_day}$ is the number of days when “bad” IEQ and “bad” productivity were reported together. Then, we calculated Spearman’s rank correlation coefficients between personality variables (N, E, O, A, C) and the ratios of self-reported IEQ, stress, productivity and co-occurring problems.

The statistical analysis of the Big Five questionnaire data and the self-reported IEQ, stress and productivity was performed with IBM SPSS Statistics 26 software. The Spearman correlation coefficient was used in the analyses.

3. Experimental results

3.1. Dataset

Of the 23 recruited office workers, data from only 15 workers were eligible for analysis as part of the sensor data were lost and some of the test subjects were unable to provide complete self-reports. The main reason for the insufficient number of obtained self-reports was the relocation of test subjects due to renovation work in the office building; 12 test subjects had to move during the study and the number of self-reports for these subjects was too small to train classifiers. Moreover, there were other gaps in the user report dataset. The test subjects did not provide reports every day, for instance, due to hurry, holidays, business trips, remote work, etc. The main reasons for sensor data losses at both

school and office were internet interruptions at the data storage site and problems reconnecting with the Azure database after the interruptions. Incidents of this type were few, but one of them caused the loss of 18 h of data, and another 6 h of data. Additionally, there was sensor data loss in the offices with wireless connection due to electromagnetic interference: in most offices 0.5%–1.5% of Bluetooth packets were lost per day, and in one office over 50% of data were lost.

As the data analysis process requires the availability of both self-reports and sensor data, the data collection problems resulted in a smaller dataset than initially planned. Furthermore, two-class classification requires sufficient data to model both classes, and some subjects never or almost never reported “bad/very bad” stress or “bad/very bad” air quality. Therefore, subjects who provided less than 10 examples of the “bad” class were excluded due to the respective classification problem. Answers to the **Overall IEQ perception** question, however, were often missing from office workers: it seems that the office worker test subjects considered it sufficient to answer only the **Air freshness** and **Odor** questions, which is understandable as complaints about stuffy air or odor obviously suggest that the air quality is bad overall. The number of examples of the “bad” class for the **Odor** question in the data appeared to be too small to train a separate classifier for detecting discomfort due to odor.

3.2. Classification of office pilot data

Data analyses were performed using the data of each target person separately, i.e. all models were person-specific. Fig. 4 illustrates the average total classification accuracies of **Any IEQ problem**, **Thermal comfort**, **Stress** and **Productivity** for office workers achieved with different sensor combinations. Here, T stands for temperature, H for humidity and P for air pressure in different sensor data combinations.

Table 3 presents the best average classification accuracies of office data for two classes of **Air freshness**, **Thermal comfort**, **Any IEQ problem**, **Stress** and **Productivity** problems, achieved with selected sensor data combinations.

The highest **Air freshness** classification accuracy with a total accuracy of 84% was achieved with the sensor data combinations of THP and THP-CO₂. The classification accuracy for **Thermal comfort** was highest (total accuracy 80%) with THP-CO₂ sensor data. Moreover, the highest accuracy for the **Any IEQ problem** (total accuracy 84%) classification was obtained when using the sensor data combinations of THP and THP-CO₂. However, it should be noted that differences in separate IEQ problems and **Any IEQ problem** classification accuracies with different sensor data combinations were rather small, ranging by only 5%. The

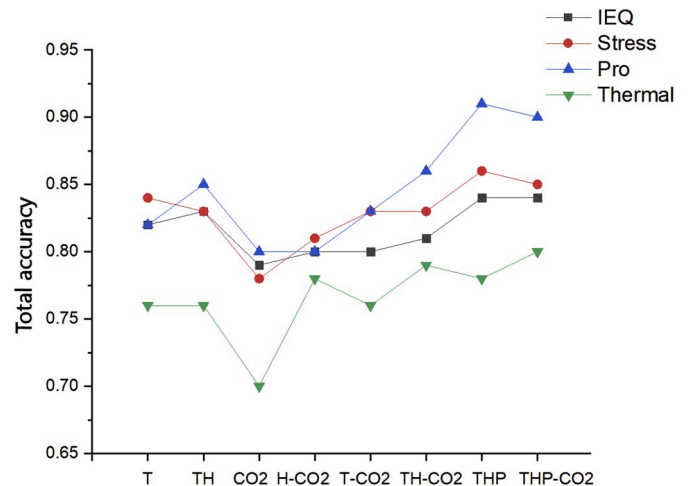


Fig. 4. Total classification accuracies of Any IEQ problem, Thermal comfort, Stress and Productivity with specific office sensor data.

Table 3

Best classification accuracies of the person-specific models for office workers.

Classification	Sensor data combination	Total accuracy	True ok	True bad	Number of subjects
Air freshness	THP	0.84	0.82	0.85	15
	THP-CO ₂	0.84	0.81	0.86	15
Thermal comfort	THP-CO ₂	0.80	0.81	0.80	12
Any IEQ problem	THP	0.84	0.86	0.82	15
	THP-CO ₂	0.84	0.86	0.81	15
Stress	THP	0.86	0.93	0.69	6
Productivity	THP	0.91	0.96	0.65	6

accuracy of **Stress** classification was highest (total accuracy 86%) with THP sensor data. For **Productivity**, the highest total accuracy of 91% was also achieved with THP sensor data. Detailed classification results with different sensor combinations are provided in [Appendix A, Table A.1](#).

[Fig. 5](#) presents the differences between IEQ, stress and productivity classification decisions and self-reported labels for two office workers for about 35 working days. The figure does not show a Y scale because the graphs are positioned along the Y axis for better visibility and because the relative positions of the values of different graphs on the same day are more important than the absolute values. To visualize and compare the ability of individual classification models to predict dissatisfaction with IEQ, stress and productivity, we chose two office workers whose number of complaints differed the most.

For comparison, the total accuracy of IEQ classification for **Office worker 1** was 0.82 (True OK = 0.84; True bad = 0.79); the total accuracy of stress classification was 0.91 (True OK = 0.89; True bad = 0.93); and the total accuracy of productivity classification was 0.82 (True OK = 0.92; True bad = 0.56). For **Office worker 2**, the total accuracy of IEQ classification was 0.81 (True OK = 0.85; True bad = 0.72); the total accuracy of stress classification was 0.79 (True OK = 0.68; True bad = 0.84); and the total accuracy for productivity classification was 0.88 (True OK = 0.97; True bad = 0.63).

3.3. Classification of school pilot data

[Fig. 6](#) presents the average total classification accuracies of **Any IEQ problem**, **Thermal comfort**, **Stress** and **Productivity** for teachers

achieved with different sensor combinations. Here, T stands for temperature, H for humidity and P for air pressure in different sensor data combinations.

[Table 4](#) presents the best average classification accuracies of the school data for two classes of **Overall IEQ perception**, **Air freshness**, **Thermal comfort**, **Any IEQ problem**, **Stress** and **Productivity**, achieved with selected sensor data combinations.

The classification accuracy of **Overall IEQ perception** was highest (total accuracy 91%) with sensor data combinations of TH, TH-CO₂ and THP. However, differences in **Overall IEQ perception** classification accuracies with different sensor data combinations were remarkably small (for details, see [Appendix A](#)). The highest **Air freshness** classification accuracy with a total accuracy of 84% was obtained with T-CO₂ sensor data. Furthermore, the accuracy of **Thermal comfort** classification was highest (total accuracy 85%) with sensor data combinations of TH-CO₂ and THP, which also classified **Overall IEQ perception** to highest accuracy best. On the other hand, complaints regarding **Overall IEQ perception** and **Thermal comfort** co-occurred frequently: on 96.5% of days when air quality discomfort was reported, thermal discomfort was also reported. For **Any IEQ problem**, the highest classification accuracy of 80% was achieved with the sensor data combinations of THP and T-CO₂. The accuracy of **Stress** classification was highest with a total accuracy of 88% when using TH sensor data and for **Productivity** with a

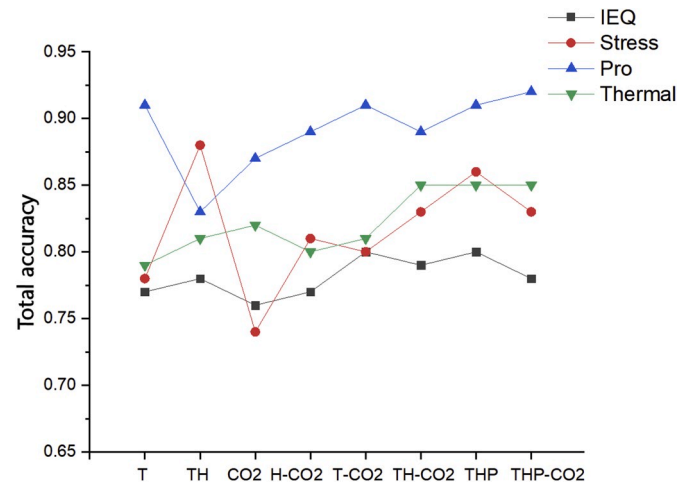


Fig. 6. Total classification accuracies of Any IEQ problem, Thermal comfort, Stress and Productivity with specific school sensor data.

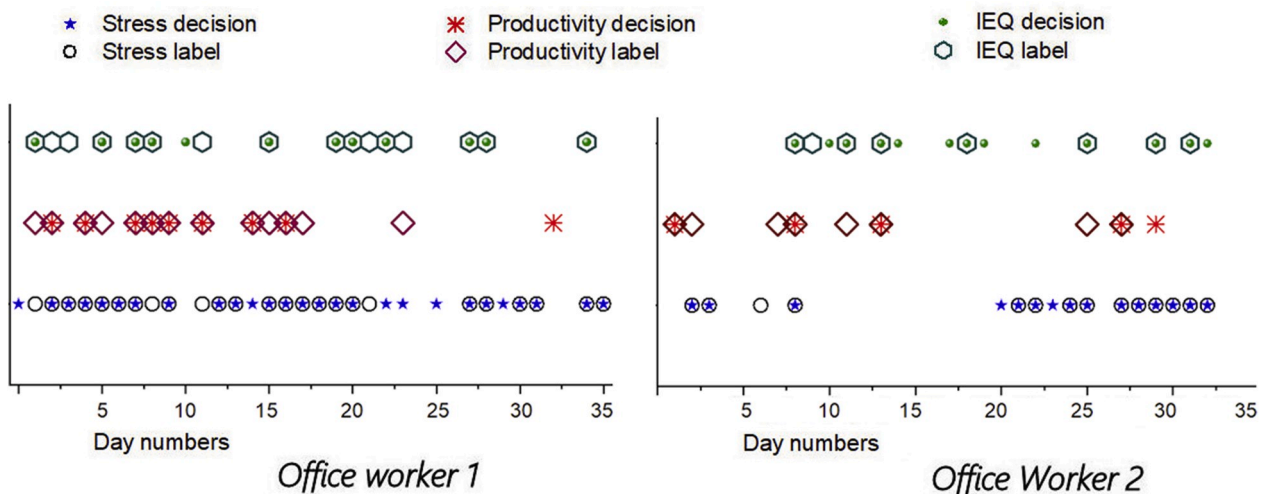


Fig. 5. Differences between IEQ, stress and productivity decisions and self-reported labels for two office workers.

Table 4

Best classification accuracies of person-specific models for elementary school teachers.

Classification	Sensor data combination	Total accuracy	True ok	True bad	Number of subjects
Overall IEQ perception	TH	0.91	0.96	0.74	4
	TH-CO ₂	0.91	0.95	0.77	4
	THP	0.91	0.96	0.74	4
Air freshness	T-CO ₂	0.84	0.85	0.82	4
Thermal comfort	TH-CO ₂	0.85	0.81	0.87	4
	THP	0.85	0.80	0.88	4
	THP-CO ₂	0.85	0.81	0.87	4
Any IEQ problem	T-CO ₂	0.80	0.76	0.83	4
	THP	0.80	0.83	0.77	4
Stress	TH	0.88	0.95	0.58	4
Productivity	THP-CO ₂	0.92	0.96	0.7	4

total accuracy of 92% when using THP-CO₂ sensor data. Compared to **IEQ** classifications, the total accuracies of the **Stress** and **Productivity** classifications varied more depending on the sensor data combinations. Detailed classification results with different sensor combinations are provided in [Appendix A, table A.2](#).

Similarly to office workers (see [Figs. 5](#)), [Fig. 7](#) presents the differences between IEQ, stress and productivity classification decisions and self-reported labels for two teachers over 35 working days. To visualize and compare the ability of individual classification models to predict dissatisfaction with IEQ, stress and productivity, we chose two teachers whose number of complaints differed the most.

For **Teacher 1** the total accuracy of IEQ classification was 0.83 (True OK = 0.56; True bad = 0.90); the total accuracy of stress classification was 0.87 (True OK = 0.64; True bad = 0.93); and the total accuracy of productivity classification was 0.73 (True OK = 0.73; True bad = 0.73). For **Teacher 2**, the total accuracy of IEQ classification was 0.73 (True OK = 0.79; True bad = 0.64); the total accuracy of stress classification was 0.71 (True OK = 0.75; True bad = 0.65); and the total accuracy of productivity classification was 0.85 (True OK = 0.90 True bad = 0.80).

3.4. Personality and differences in perceptions of IEQ, stress and productivity

We also investigated correlations between five personality variables (N, E, O, A, C) and the ratios of different “bad” self-reports (see Equations (4)–(9)) for 16 test subjects who responded to the Big Five questionnaire. Spearman’s rank correlation coefficients between personality

variables (N, E, O, A, C) and the ratios of self-reported IEQ problems, stress and productivity are shown in [Table 5](#). Here r and P denote Spearman’s rank correlation coefficient and probability value, respectively.

4. Discussion

Modern buildings need to be designed to fulfill various external, financial, regulatory, and technical requirements, while providing a comfortable environment for the occupants. This study increases our understanding of how to better address the individual human characteristics of building use by assessing their significance for the wellbeing and performance of the individuals and, ultimately, the organization operating within a building. Despite the tendency to prioritize hard requirements among construction design stakeholders [4], the authors would like to highlight the significance of human factors.

Employee expenses significantly exceed building operation and rental costs [7,13], and the well-being of employees has emerged as a strategic priority of enterprises [5]. Employee satisfaction and well-being are typically assessed using periodic surveys; however, these are conducted infrequently, if at all, and employees might resign or develop burnout between surveys. However, the reasons behind stress-related reduction in employee well-being often remain unknown to the management, and the same problems may arise with new employees. Therefore, more frequent but less burdensome means of assessing employee satisfaction and well-being are needed.

The main contribution of this study is to use environmental sensors to assess the perceived IEQ, stress and productivity of teachers and office workers during their normal daily work routine in real buildings. Furthermore, our data collection period varied in duration from three to seven months for different test subjects, which is long-term compared to the average duration reported in related laboratory and real-life studies [8]. Previous studies have emphasized the importance of real-life experimental setups and acquiring quantitative data due to observable differences between actual authentic operating environments versus artificially arranged conditions [15,24]. For classifying IEQ, stress and productivity, we used SVM classifier, which is one of the most successful classifier types in stress detection to date [17,24,34]. Our data analysis indicated that relatively high accuracies can be achieved using person-specific SVM models to classify self-reported IEQ, stress and productivity into “negative” and “positive” classes.

4.1. IEQ classification

The experiment was conducted for two different environment types: school and office. The measurement framework was able to provide valid data from both spaces. Despite the differences in the type of usage

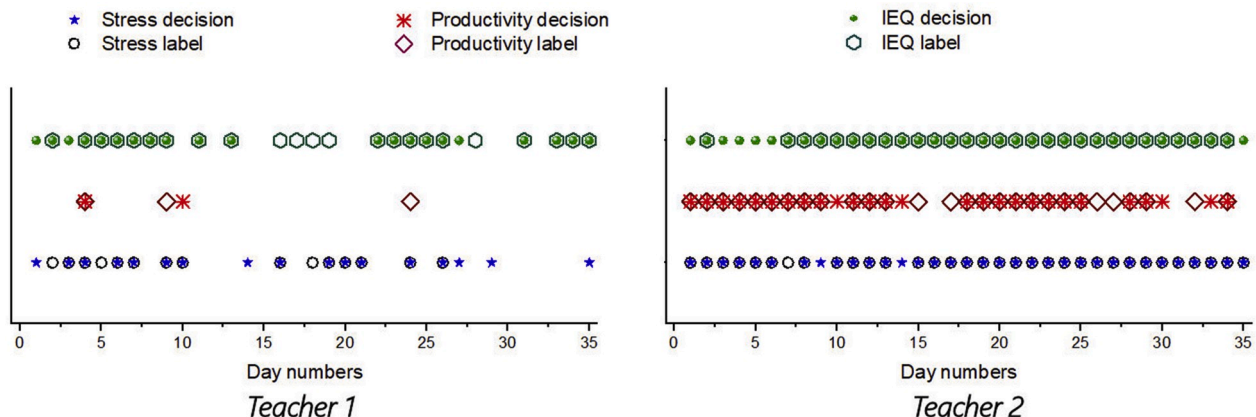


Fig. 7. Differences between IEQ, stress and productivity decisions and self-reported labels for two teachers.

Table 5

Spearman's correlations (2-tailed) in relation to perceived IEQ, stress, productivity and co-occurring ratios.

Variable, r value; P value	Neuroticism	Extroversion	Openness	Agreeableness	Conscientiousness
Stress ratio	−0.207; 0.440	0.349; 0.184	0.407; 0.120	0.237; 0.377	0.184; 0.495
Productivity ratio	0.079; 0.771	0.155; 0.566	0.114; 0.671	−0.409; 0.116	−0.251; 0.349
IEQ ratio	−0.605; 0.013	0.687; 0.003	0.483; 0.058	0.156; 0.562	0.005; 0.986
Stress Productivity ratio	−0.264; 0.327	0.439; 0.089	0.400; 0.125	−0.027; 0.920	−0.065; 0.810
IEQ Stress ratio	−0.344; 0.192	0.720^a; 0.00167	0.480; 0.060	−0.125; 0.644	−0.183; 0.498
IEQ Productivity ratio	−0.493; 0.052	0.507; 0.045	0.172; 0.525	0.300; 0.260	−0.059; 0.827

Spearman's correlation analysis revealed a positive correlation between extroversion IEQ Stress ratio ($r = 0.72$; $n = 16$, $P = 0.00167$, 2-tailed), which achieved a statistical threshold adjusted for a sample size of 16.

^a Statistically significant when $P < 0.0017$.

and configuration, the number of typical occupant analysis results was quite aligned. However, regarding the sensor set applied, in the school environment it appears that all of the selected sensors or combinations provided almost equally accurate results, whereas in the office environment a more complex set of sensors, such as THP or THP-CO₂, enabled best analysis of the results compared to any of the single sensors alone. The presence of an air pressure (P) sensor seemed to improve the accuracy in both scenarios. If selecting a set of sensors based on these results, more weight should be given to the office scenario, due to the larger amount of participants.

Even though the occupants may have had varying expectations regarding the two different spaces, based on the results, no major differences were found between office worker and teacher perceptions. With the optimal sensor set, the average accuracy of **Overall IEQ perception** for teachers was 91%. For **Air freshness**, the average classification accuracy was 84% and for **Thermal comfort** 80% for office workers and 91% for teachers. However, dissatisfaction with overall air quality and thermal comfort often co-occurred in the school environment: “bad” **Overall IEQ perception** was reported on the same day in 96.5% of cases when “bad” **Thermal comfort** was reported. This probably this reflects the changes in different environmental variables due to rapid changes in the number of pupils per classroom. For **Any IEQ problems**, the best average classification accuracy was 84% for office workers and 80% for teachers. To summarize, these findings suggest that different combinations of fairly inexpensive environmental sensors can be used to assess occupant environmental comfort in different types of indoor environments to a level of accuracy sufficient for decision making. To highlight one, the THP set of sensors appears to lead to best classification accuracy.

Direct comparison of our results with other related works is not possible due to different setups, datasets and evaluation approaches. Cheung et al. [27] monitored 15 test subjects in a 7-day experiment where a portable sensor device measured temperature, relative humidity and CO₂ concentration, but found only a weak association between self-reported thermal and air quality comfort and objective IEQ measures. Moreover, IEQ perceptions are not only affected by environmental parameters per se, but also by personal expectations, environmental context, availability of control and various individual variables (e.g. Ref. [26,27]), which makes comparison challenging.

4.2. Stress and productivity classification

The total average accuracy of stress detection with the best sensor set was 86% for office workers and 88% for teachers. The best stress classification accuracy was achieved for office workers with THP sensors (total accuracy 86%) and for teachers with TH sensors (total accuracy 88%). Previous studies employing supervised learning and real-life mobile data for stress classification have reported accuracies ranging from 60% to 75% [17,20,21]. Sano et al. [20] reported 64% accuracy when stress was recognized on the basis of mobile phone data only, and 78% accuracy when data from two wearables were used for stress recognition. Reported accuracies for stress detection based on physiological sensors have varied considerably, and experiments with larger

datasets have resulted in lower accuracies. For instance, a study with 1000 subjects reported that for 38% of test subjects the stress recognition accuracy was as good as random guessing [46]. The stress classification accuracy achieved in this experiment with 2 target categories is higher, but not directly comparable with previous studies with 3 or more target classes. However, compared to mobile phone [18,21], audio [22] or keyboard usage based stress detection, environmental sensors are more privacy safe. As regards perceived productivity, the average person-specific classification accuracy was 91% for office workers and 92% for teachers. The best sensor data combination for work productivity classification seemed to be THP for office workers and THP-CO₂ for teachers. This finding is consistent with previous studies showing that both thermal and indoor air quality discomfort have negative effects on work performance [8,12], and productivity [13].

4.3. Individual differences

Perceptions of environment, stress and productivity are highly individual experiences, and the test subjects of this study reported different numbers of IEQ, stress and productivity problems (see Figs. 5 and 7). Since personality is known to be associated with individual experience of stressors [30], we were also interested in studying how personality traits are associated with perceived IEQ, stress and productivity. Spearman's correlation analysis revealed a correlation between extroversion and IEQ Stress ratio ($r = 0.72$; $n = 16$; $P = 0.00167$; 2-tailed), which achieved a statistical threshold adjusted for a sample size of 16. This finding indicates that more extroverted individuals are more stressed by insufficient environmental quality or become more sensitive to their environment when experiencing stress. This is interesting, as high scores of extroversion are typically associated with positive feelings [47]. Interestingly, we did not find a correlation between neuroticism and self-reported stress, even though there is a known association between these [29], and agreeableness and conscientiousness did not correlate with any variables. On the other hand, our data sample was small and it may be that personality has only a minor effect on short-term actions, whereas other individual variables, thoughts, mood and context may have bigger effects on perceptions than personality [12]. Nevertheless, the finding is of interest for building design stakeholders and worthy of further study, opening up the idea of personality-based individual tailoring and modification of personal spaces.

4.4. Limitations and suggestions for future work

The main limitation of this work is the relatively small number of test subjects due to real-world external events that reduced the number of eligible participants. However, the average number of answers per test subject in our data was 75. Such a data size suffices only to train a 2-class model. However, this is a rather typical data size for many studies on stress detection: for example, in Ref. [17,18] the data size was 50 and 65 self-reports per test subject on average, respectively. The initial results indicate that it would be interesting to extend the scale for wider evaluation of personality factors influencing individual perception of

environmental quality, and even suggest that personal spaces could be designed based on personality categories. Another limitation is that we did not evaluate stress and productivity objectively, considering the realworld experiment setup versus the requirements for conducting objective stress measurement over the long term. Subjective perceptions of productivity have been clearly shown to not consistently correspond to objective measurements of productivity [8]. On the other hand, maximum productivity greatly varies among individuals, and personal opinion can thus be considered as sufficient data within the context of this study. Moreover, the causal effects of insufficient IEQ on health and productivity are difficult to study in real environments because multiple other human related factors, such as the social and organizational environment, may affect occupant health and productivity [48]. As this study was conducted in a small city in northern Finland where the outside air is typically high quality, the authors do not consider participants' response bias due to external pollution exposure during transit or otherwise before the place of measurement to be an influencing factor. It should be noted, however, that in large cities or areas impacted by pollution, external air quality might impact the perceptions of participants in similar studies.

Our future work will focus on the interaction between people, their behavioral patterns and the working environment at both the individual and organizational level. We consider the results of this study to be a good starting point for further studies into using real-life objective sensor data for improving the satisfaction and wellbeing of employees and organizations. We plan to collect a larger amount of environmental data over a long-term period to validate our results. We will also continue our data analysis with a view to providing personal suggestions or environmental adjustments based on different working tasks and individual needs, while combining these with building energy efficiency improvement work. Furthermore, our aim is to develop an organizational barometer to assess long-term wellbeing and stress, not only at the individual level, but also at the organizational level, in order to measure the human wellbeing of the whole organization frequently and on a continuous basis. Providing more precise information could enable enhanced decision-making to address the human resources needs of organizations.

5. Conclusions

Healthy, motivated and skillful employees are a key success factor for enterprises [5]. Insufficient environmental quality can strain employees' health and wellbeing, leading to productivity losses. This paper

introduced a method to assess employees' perceptions of IEQ, stress and productivity using environmental sensor data and machine learning. The results indicated that objective IEQ measures can classify self-reported IEQ (84%), stress (88%) and productivity (92%) using a person-specific SVM classifier. By comparing various sensor combinations, a recommended optimal set of sensors was found for each target class.

According to our findings, individuals of a more extroverted personality type are more likely to be stressed by poor environmental quality or to be more sensitive to environmental factors when experiencing stress compared to less extroverted individuals. As a future line of research, the goal of personality-based individual tailoring and modification of personal spaces was presented. Our findings indicate that by applying low-cost commercial sensors, it is feasible to measure and classify indoor environment quality sufficiently accurately for comprehensive decision making. Environmental quality seems to have a relation to perceived individual stress and productivity. As a future research direction, the authors propose a continuous barometer based on sensor measurements and self-reports to address the entire organization's current, past, and even predicted stress and productivity status, for the improved management of human resource wellbeing.

Declarations of competing interest

None to declare.

Acknowledgements

The authors are grateful to the pilot participants for their active and significant role in both pilots. Special thanks to Mr. Severi Olsbo and our former colleague Mr. Dan Bendas for their participation in the implementation of the data collection solution. We are also thankful to Dr. Panu Korpipää for valuable comments. The study was implemented in collaboration with the ESTABLISH and SCOTT projects. ESTABLISH (ITEA3 15008) was supported by Business Finland and VTT Technical Research Centre of Finland, and the school pilot was designed in collaboration with UniqAir, CGI, and InspectorSec. SCOTT (Secure COnnected Trustable Things) received funding from the Electronic Component Systems for European Leadership Joint Undertaking under grant agreement No 737422. This Joint Undertaking receives support from the European Union's Horizon 2020 research and innovation program, and from Austria, Spain, Finland, Ireland, Sweden, Germany, Poland, Portugal, Netherlands, Belgium and Norway.

Appendix A. Classification accuracies of person specific models with different sensor data combinations

Table A.1

Classification accuracies of person-specific models for office workers with different sensor data combinations.

Classification	Sensor data combination	Total accuracy	True ok	True bad	Number of subjects
Air freshness	T	0.81	0.79	0.83	15
	TH	0.83	0.82	0.83	15
	CO ₂	0.79	0.79	0.80	15
	H-CO ₂	0.79	0.76	0.82	15
	T-CO ₂	0.80	0.77	0.83	15
	TH-CO ₂	0.82	0.79	0.85	15
	THP	0.84	0.82	0.85	15
	THP-CO ₂	0.84	0.81	0.86	15
Thermal comfort	T	0.76	0.73	0.79	10
	TH	0.76	0.75	0.77	10
	CO ₂	0.70	0.69	0.70	12
	H-CO ₂	0.78	0.77	0.79	12
	T-CO ₂	0.76	0.74	0.79	12
	TH-CO ₂	0.79	0.77	0.80	12

(continued on next page)

Table A.1 (continued)

	THP	0.78	0.75	0.80	10
	THP-CO₂	0.80	0.81	0.80	12
Any IEQ problem	T	0.82	0.84	0.79	15
	TH	0.83	0.84	0.81	15
	CO ₂	0.79	0.80	0.78	15
	H-CO ₂	0.80	0.83	0.77	15
	T-CO ₂	0.80	0.83	0.76	15
	TH-CO ₂	0.81	0.84	0.78	15
	THP	0.84	0.86	0.82	15
	THP-CO₂	0.84	0.86	0.81	15
Stress	T	0.84	0.90	0.70	6
	TH	0.83	0.90	0.64	6
	CO ₂	0.78	0.86	0.61	8
	H-CO ₂	0.81	0.87	0.69	8
	T-CO ₂	0.83	0.90	0.69	8
	TH-CO ₂	0.83	0.90	0.68	8
	THP	0.86	0.93	0.69	6
	THP-CO₂	0.85	0.92	0.7	6
Productivity	T	0.82	0.93	0.34	6
	TH	0.85	0.95	0.42	6
	CO ₂	0.80	0.93	0.25	8
	H-CO ₂	0.80	0.90	0.38	8
	T-CO ₂	0.83	0.90	0.69	8
	TH-CO ₂	0.86	0.94	0.51	8
	THP	0.91	0.96	0.65	6
	THP-CO₂	0.90	0.97	0.61	8

Table A.2

Classification accuracies of person-specific models for elementary school teachers with different sensor data combinations

Classification	Sensor data combination	Total accuracy	True ok	True bad	Number of subjects
Overall IEQ perception	T	0.90	0.95	0.73	4
	TH	0.91	0.96	0.74	4
	CO ₂	0.88	0.94	0.70	4
	H-CO ₂	0.90	0.95	0.72	4
	T-CO ₂	0.90	0.95	0.76	4
	TH-CO₂	0.91	0.95	0.77	4
	THP	0.91	0.96	0.74	4
	THP-CO₂	0.90	0.95	0.72	4
Air freshness	T	0.81	0.80	0.82	4
	TH	0.78	0.75	0.80	4
	CO ₂	0.76	0.76	0.76	4
	H-CO ₂	0.74	0.74	0.74	4
	T-CO₂	0.84	0.85	0.82	4
	TH-CO ₂	0.80	0.81	0.79	4
	THP	0.78	0.75	0.81	4
	THP-CO₂	0.80	0.80	0.79	4
Thermal Comfort	T	0.79	0.76	0.82	4
	TH	0.81	0.77	0.84	4
	CO ₂	0.82	0.80	0.84	4
	H-CO ₂	0.80	0.78	0.81	4
	T-CO ₂	0.81	0.79	0.84	4
	TH-CO₂	0.85	0.81	0.87	4
	THP	0.85	0.80	0.88	4
	THP-CO₂	0.85	0.81	0.87	4
Any IEQ problem	T	0.77	0.78	0.75	4
	TH	0.78	0.78	0.78	4
	CO ₂	0.76	0.74	0.78	4
	H-CO ₂	0.77	0.74	0.79	4
	T-CO₂	0.80	0.76	0.83	4
	TH-CO ₂	0.79	0.76	0.81	4
	THP	0.80	0.83	0.77	4
	THP-CO₂	0.78	0.76	0.79	4
Stress	T	0.78	0.83	0.69	4
	TH	0.88	0.95	0.58	4
	CO ₂	0.74	0.83	0.57	4
	H-CO ₂	0.81	0.89	0.66	4

(continued on next page)

Table A.2 (continued)

Classification	Sensor data combination	Total accuracy	True ok	True bad	Number of subjects
	T-CO ₂	0.80	0.86	0.69	4
	TH-CO ₂	0.83	0.89	0.71	4
	THP	0.86	0.89	0.79	4
	THP-CO ₂	0.83	0.89	0.74	4
Productivity	T	0.91	0.96	0.69	4
	TH	0.83	0.91	0.69	4
	CO ₂	0.87	0.92	0.68	4
	H-CO ₂	0.89	0.94	0.68	4
	T-CO ₂	0.91	0.95	0.73	4
	TH-CO ₂	0.89	0.94	0.68	4
	THP	0.91	0.96	0.67	4
	THP-CO ₂	0.92	0.96	0.7	4

References

- [1] A. Steinemann, P. Wargocki, B. Rismanchi, Ten questions concerning green buildings and indoor air quality, *Build. Environ.* 112 (2017) 351–358.
- [2] W.J. Fisk, Health and productivity gains from better indoor environments and their relationship with building energy efficiency, *Annu. Rev. Energy Environ.* 25 (1) (2000) 537–566.
- [3] J.G. Allen, P. MacNaughton, U. Satish, S. Santanam, J. Vallarino, J.D. Spengler, Associations of cognitive function scores with carbon dioxide, ventilation, and volatile organic compound exposures in office workers: a controlled exposure study of green and conventional office environments, *Environ. Health Perspect.* 124 (6) (2015) 805–812.
- [4] M. Jia, A. Komeily, Y. Wang, R.S. Srinivasan, Adopting Internet of Things for the development of smart buildings: a review of enabling technologies and applications, *Autom. Constr.* 101 (2019) 111–126.
- [5] Deloitte, 2018. <https://www2.deloitte.com/content/dam/insights/us/articles/HCTrends2018/2018-HCTrendsRise-of-the-social-enterprise.pdf>. last accessed 2020/02/01.
- [6] D. Clements-Croome (Ed.), *Creating the Productive Workplace*, Taylor & Francis, 2006.
- [7] J. Alker, M. Malanca, C. Pottage, R. O'Brien, Health, Wellbeing & Productivity in Offices: the Next Chapter for Green Building, World Green Building Council, 2014.
- [8] P. Wargocki, D.P. Wyon, Ten questions concerning thermal and indoor air quality effects on the performance of office work and schoolwork, *Build. Environ.* 112 (2017) 359–366.
- [9] O. Seppänen, P. Wargocki, J. Andersson, A. Boerstra, D. Clements-Croome, K. Fitzner, S. Hanssen, *Indoor Climate and Productivity in Offices*, REHVA Guidebook, 2006 (6).
- [10] S. Deng, J. Lau, Seasonal variations of indoor air quality and thermal conditions and their correlations in 220 classrooms in the Midwestern United States, *Build. Environ.* 157 (2019) 79–88.
- [11] T. Parkinson, A. Parkinson, R. de Dear, Continuous IEQ monitoring system: context and development, *Build. Environ.* 149 (2019) 15–25.
- [12] S. Lamb, K.C. Kwok, A longitudinal investigation of work environment stressors on the performance and wellbeing of office workers, *Appl. Ergon.* 52 (2016) 104–111.
- [13] Y. Al Horr, M. Arif, A. Kaushik, A. Mazroei, M. Katafygiotou, E. Elsarrag, Occupant productivity and office indoor environment quality: a review of the literature, *Build. Environ.* 105 (2016) 369–389.
- [14] Y. Geng, W. Ji, B. Lin, Y. Zhu, The impact of thermal environment on occupant IEQ perception and productivity, *Build. Environ.* 121 (2017) 158–167.
- [15] P. Kumar, A.N. Skouloudis, M. Bell, M. Viana, M.C. Carotta, G. Biskos, L. Morawska, Real-time sensors for indoor air monitoring and challenges ahead in deploying them to urban buildings, *Sci. Total Environ.* 560 (2016) 150–159.
- [16] K. Plarre, A. Raji, S.M. Hossain, A.A. Ali, M. Nakajima, M. Al'Absi, et al., April). Continuous inference of psychological stress from sensory measurements collected in the natural environment, in: *Proceedings of the 10th ACM/IEEE International Conference on Information Processing in Sensor Networks*, IEEE, 2011, pp. 97–108.
- [17] A. Maxhuni, P. Hernandez-Leal, L.E. Sucar, V. Osmani, E.F. Morales, O. Mayora, Stress modelling and prediction in presence of scarce data, *J. Biomed. Inf.* 63 (2016) 344–356.
- [18] M. Ciman, K. Wac, Individuals' stress assessment using human-smartphone interaction analysis, *IEEE Trans. Affect. Comput.* 9 (1) (2016) 51–65.
- [19] E. Vildjiounaite, J. Kallio, V. Kyllönen, M. Nieminen, I. Määtänen, M. Lindholm, et al., Unobtrusive stress detection on the basis of smartphone usage data, *Personal Ubiquitous Comput.* 22 (4) (2018) 671–688.
- [20] A. Sano, S. Taylor, A.W. McHill, A.J. Phillips, L.K. Barger, E. Klerman, R. Picard, Identifying objective physiological markers and modifiable behaviors for self-reported stress and mental health status using wearable sensors and mobile phones: observational study, *J. Med. Internet Res.* 20 (6) (2018), e210.
- [21] A. Muaremi, B. Arnrich, G. Tröster, Towards measuring stress with smartphones and wearable devices during workday and sleep, *BioNanoScience* 3 (2) (2013) 172–183.
- [22] J. Hernandez, R.R. Morris, R.W. Picard, Call center stress recognition with person-specific models, in: *International Conference on Affective Computing and Intelligent Interaction*, Springer, Berlin, Heidelberg, 2011, October, pp. 125–134.
- [23] D. Carneiro, P. Novais, J.M. Pêgo, N. Sousa, J. Neves, June). Using mouse dynamics to assess stress during online exams, in: *International Conference on Hybrid Artificial Intelligence Systems*, Springer, Cham, 2015, pp. 345–356.
- [24] E. Vildjiounaite, V. Huotari, J. Kallio, V. Kyllönen, S.M. Mäkelä, G. Gimel'farb, Unobtrusive Assessment of Stress of Office Workers via Analysis of Their Motion Trajectories, *Pervasive and Mobile Computing*, 2019.
- [25] J. Kallio, E. Vildjiounaite, V. Kyllönen, J. Ronkainen, J. Koivusaari, S. Muuraikangas, et al., Classifying teachers' self-reported productivity, stress and indoor environmental quality using environmental sensors, in: *European Conference on Ambient Intelligence*, Springer, Cham, 2019, November, pp. 27–40.
- [26] M. Frontczak, P. Wargocki, Literature survey on how different factors influence human comfort in indoor environments, *Build. Environ.* 46 (4) (2011) 922–937.
- [27] T.C. Cheung, S. Schiavon, E.T. Gall, M. Jin, W.W. Nazaroff, Longitudinal assessment of thermal and perceived air quality acceptability in relation to temperature, humidity, and CO₂ exposure in Singapore, *Build. Environ.* 115 (2017) 80–90.
- [28] N. Bolger, A. Zuckerman, A framework for studying personality in the stress process, *J. Pers. Soc. Psychol.* 69 (5) (1995) 890.
- [29] L.R. Goldberg, An alternative "description of personality": the big-five factor structure, *J. Pers. Soc. Psychol.* 59 (6) (1990) 1216.
- [30] G.P. Chrousos, Stress and disorders of the stress system, *Nat. Rev. Endocrinol.* 5 (7) (2009) 374.
- [31] L. Lan, P. Wargocki, Z. Lian, Quantitative measurement of productivity loss due to thermal discomfort, *Energy Build.* 43 (5) (2011) 1057–1062.
- [32] J. Aigrain, S. Dubuisson, M. Detyniecki, M. Chetouani, May). Person-specific behavioral features for automatic stress detection, in: *2015 11th IEEE International Conference and Workshops on Automatic Face and Gesture Recognition (FG)*, vol. 3, IEEE, 2015, pp. 1–6.
- [33] S. Koldijk, M.A. Neerinx, W. Kraaij, Detecting work stress in offices by combining unobtrusive sensors, *IEEE Trans. Affect. Comput.* 9 (2) (2016) 227–239.
- [34] A. Alberdi, A. Aztiria, A. Basarab, Towards an automatic early stress recognition system for office environments based on multimodal measurements: a review, *J. Biomed. Inf.* 59 (2016) 49–75.
- [35] Y. Peng, Z. Nagy, A. Schlüter, Temperature-preference learning with neural networks for occupant-centric building indoor climate controls, *Build. Environ.* 154 (2019) 296–308.
- [36] T. Chaudhuri, D. Zhai, Y.C. Soh, H. Li, L. Xie, Thermal comfort prediction using normalized skin temperature in a uniform built environment, *Energy Build.* 159 (2018) 426–440.
- [37] A. Aryal, B. Becerik-Gerber, A Comparative Study of Predicting Individual Thermal Sensation and Satisfaction Using Wrist-Worn Temperature Sensor, Thermal Camera and Ambient Temperature Sensor, *Building and Environment*, 2019, p. 106223.
- [38] <http://www.efanet.org/air-quality/indoor-air-quality> last accessed 2020/02/01.
- [39] C.J. Soto, O.P. John, Short and extra-short forms of the big five inventory-2: the BFI-2-S and BFI-2-XS, *J. Res. Pers.* 68 (2017) 69–81.
- [40] H. Similä, S. Muuraikangas, J. Ronkainen, K. Vehmas, J. Kallio, Design and feasibility evaluation of self-reporting application for indoor air quality and health measures, in: *Multidisciplinary Digital Publishing Institute Proceedings*, vol. 31, 2019, p. 47, 1.
- [41] https://www.vtt.fi/files/events/Teollinen_Internet_ja_Digitalisaatio_2015/TinyNode.pdf?search=tiny%20node last accessed 2020/02/01.
- [42] Finnish Ministry of Social Affairs and Health: Decree of the Ministry of Social Affairs and Health on Health-Related Conditions of Housing and Other Residential Buildings and Qualification Requirements for Third-Party Experts (545/2015). <https://www.finlex.fi/en/laki/kaannokset/2015/en20150545.pdf> last accessed 2020/02/01.
- [43] <https://docs.microsoft.com/en-us/python/api/azure-cosmosdb-table/azure.cosmosdb.table.tableservice?view=azure-python> last accessed 2020/02/01.
- [44] <https://scikit-learn.org/> last accessed 2020/02/01.
- [45] <https://buildmedia.readthedocs.org/media/pdf/tsfresh/latest/tsfresh.pdf> last accessed 2020/02/01.

- [46] E. Smets, E.R. Velazquez, G. Schiavone, I. Chakroun, E. D'Hondt, W. De Raedt, et al., Large-scale wearable data reveal digital phenotypes for daily-life stress detection, *npj Digit. Med.* 1 (1) (2018) 67.
- [47] E. Komulainen, K. Meskanen, J. Lipsanen, J.M. Lahti, P. Jylhä, T. Melartin, et al., The effect of personality on daily life emotional processes, *PloS One* 9 (10) (2014), e110907.
- [48] C. Brauer, S. Mikkelsen, The influence of individual and contextual psychosocial work factors on the perception of the indoor environment at work: a multilevel analysis, *Int. Arch. Occup. Environ. Health* 83 (6) (2010) 639–651.